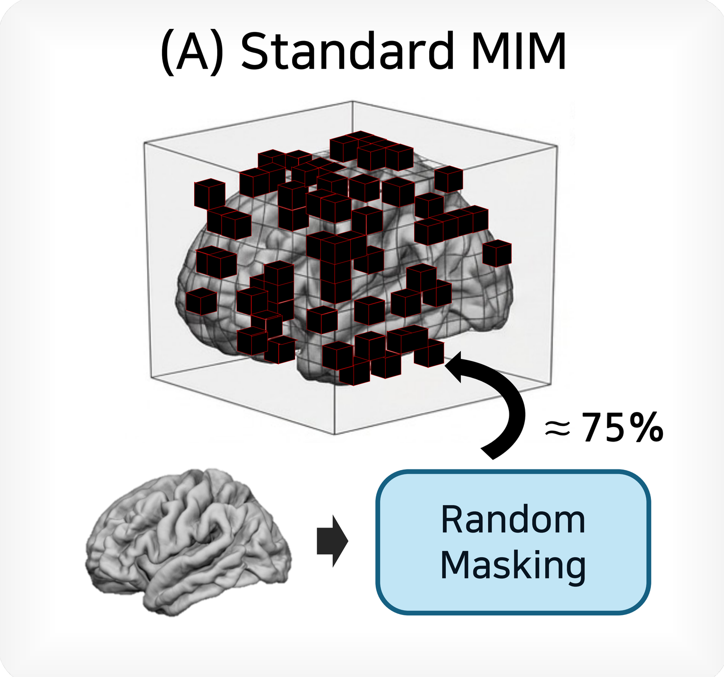
Anatomy-Aware Masked Image Modeling for Self-Supervised Learning on 3D Brain MRI

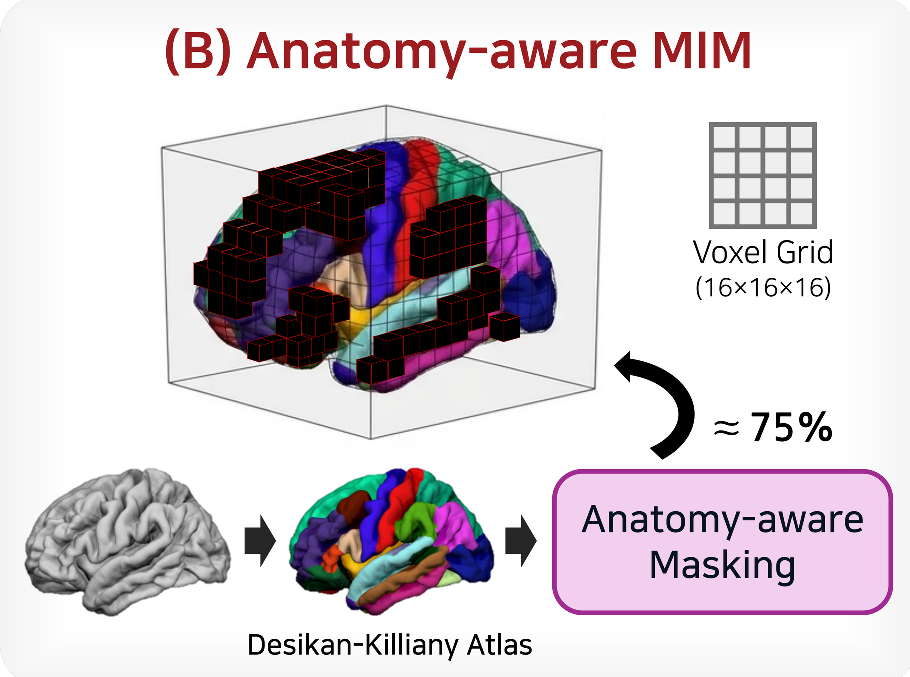
Yeonwoo Kim and Won Hee Lee
Department of Software Convergence, Kyung Hee University
{dusdn,whlee}@khu.ac.kr

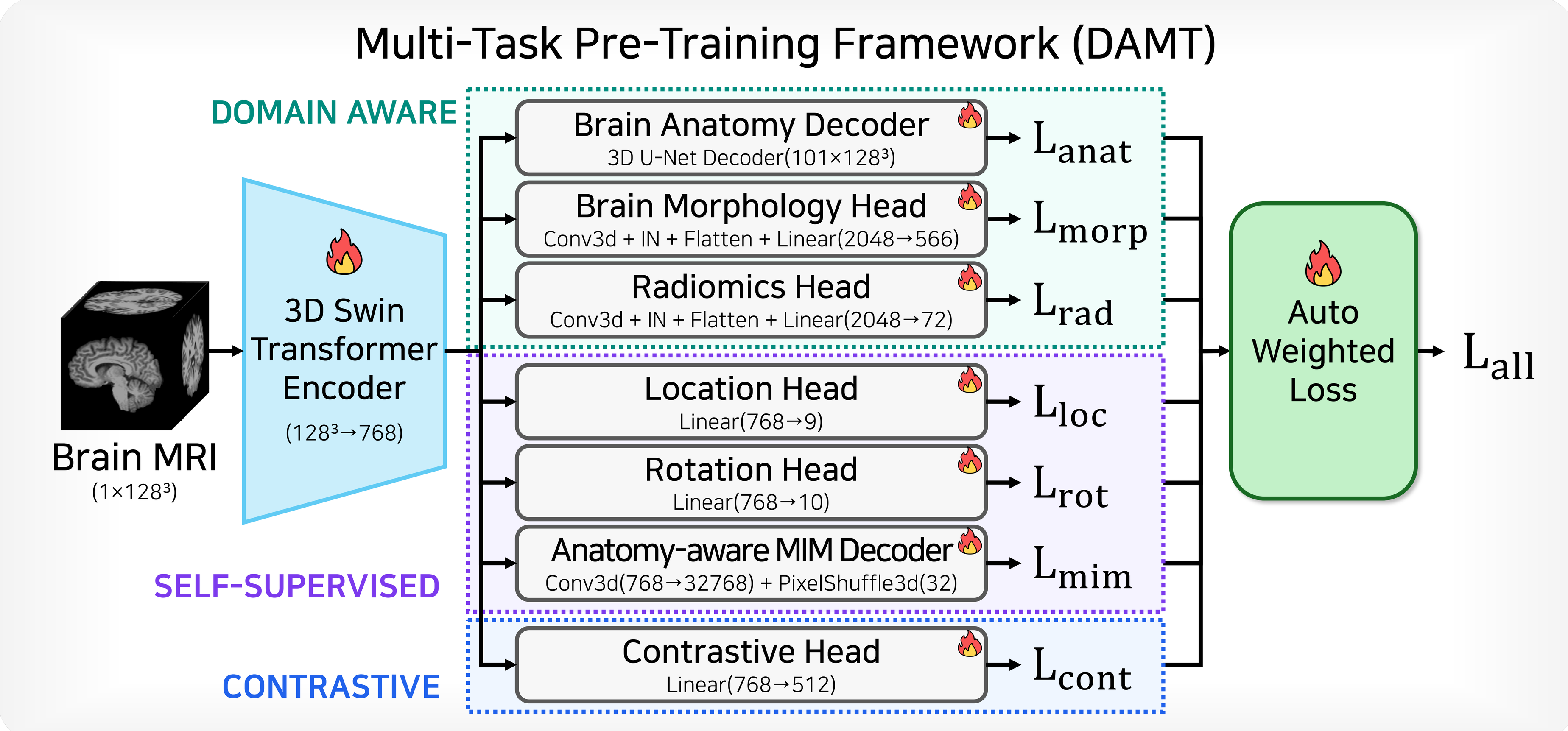
Introduction

- Self-supervised learning (SSL) via masked image modeling (MIM) effectively learns transferable brain representations from large-scale unlabeled neuroimaging datasets [1].
- However, standard MIM uniformly masks random patches, disregarding the brain's intrinsic anatomical organization [2] and rewarding local texture matching rather than clinically relevant global structural reasoning.
- This study proposes **Anatomy-aware MIM**, a novel masking strategy for 3D brain MRI that utilizes region-level masking derived from cortical parcellations.
- By masking entire anatomically coherent regions at once, the framework forces the encoder to reconstruct missing brain structures using long-range inter-regional context, providing a meaningful inductive bias for neuropsychiatric conditions.

Methods







- **Standard MIM (A)** masks 75% of non-overlapping 16^3-voxel patches uniformly at random. Encourages models to interpolate missing values from immediately neighboring patches, bypassing underlying anatomical logic.
- **Anatomy-aware MIM (B)** assigns each 16^3 patch a dominant anatomical label based on the Desikan–Killiany cortical atlas [3]. Groups patches by anatomical region and masks entire regions simultaneously until reaching the ≈75% target ratio. Forces the model to reconstruct missing structures from spatially contiguous, long-range structural context.
- The multi-task pre-training framework [4] employs a 3D Swin Transformer [5] encoder to process T1-weighted brain MRI volumes of size 128^3voxels at 1.25 mm isotropic resolution.
- During pre-training, the model jointly optimizes seven complementary pretext tasks.
- To efficiently handle competing gradient signals, loss weights across these tasks are automatically learned via uncertainty-based weighting [6], eliminating the need for manual tuning.

Experiments

- The model was pre-trained on 15,532 T1w brain MRIs and evaluated on 2,050 subjects (HC: 1,460, SCZ: 590) across multiple datasets [7-13].
- Pre-training was conducted for 100 epochs using the AdamW optimizer with a 5×10^{-4} learning rate and cosine annealing.
- Downstream performance was evaluated via 5-fold cross-validation linear probing targeting schizophrenia classification.

Results

Metrics	Random MIM	Ours
Balanced Accuracy	0.647	**0.669**
AUC	0.767	**0.771**
F1-Score	0.482	**0.522**
Precision	0.609	**0.619**
Recall	0.398	**0.451**

- Anatomy-aware MIM consistently outperformed random masking in schizophrenia classification across all metrics.
- Achieved notable improvements in balanced accuracy (64.7% to 66.9%), AUC (0.767 to 0.771), and a clinically valuable 13.3% relative increase in Recall (0.398 to 0.451).

Discussions

- Anatomy-aware MIM yields consistent improvements in schizophrenia classification, demonstrating that self-supervised objectives should reflect the intrinsic anatomical structure of the brain.
- The notable increase in recall is particularly meaningful, as it shows the model effectively captures the distributed inter-regional morphological alterations characteristic of schizophrenia.
- Ultimately, these gains underscore the value of directly embedding neuroanatomical priors into pre-training through a simple modification that requires no additional parameters or annotations.

Acknowledgment

This work was supported in part by the Institute of Information & Communications Technology Planning & Evaluation (IITP), funded by the Korea government (MSIT), under grants RS-2024-00509257 (Global AI Frontier Lab) and IITP-2026-RS-2024-00438239 (ITRC); in part by the National Research Foundation of Korea (NRF), funded by MSIT (RS-2026-25485605); and in part by the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health and Welfare (RS-2025-02293110).

References

[1] He et al., *CVPR*, 2022.
[2] van Erp et al., *Mol. Psychiatry*, 2016.
[3] Desikan et al., *NeuroImage*, 2006.
[4] Kim et al., *ACCV*, 2024.
[5] Liu et al., *ICCV*, 2021.
[6] Kendall et al., *CVPR*, 2018.
[7] Cerri et al., arXiv, 2025.
[8] Kennedy et al., *Neuroinformatics*, 2012.
[9] Repovš et al, *Front. Hum. Neurosci.*, 2012.
[10] Aine et al., *Neuroinformatics*, 2017.
[11] Tanaka et al., *Sci. Data*, 2021.
[12] Gollub et al., *Neuroinformatics*, 2013.
[13] Wang et al., *Front. Neuroinform.*, 2013.

Scan the QR code for more information!

Access to the Paper